

## 19.6 Hermitian Perturbations

For Hermitian matrices, Chapter 16 already gave the basic perturbation bound: each eigenvalue moves by at most  $\|\mathbf{E}\|_2$ , and no smaller constant is possible. This section asks what that bound does **not** tell us, and answers two such questions. The first is practical. A computation rarely hands us  $\mathbf{E}$ ; it hands us an approximate eigenvector, and the question is how good the eigenvalue estimate it yields is. The answer is a residual bound, and when the eigenvalue is separated from the others it improves from linear to **quadratic** in the residual; that is the Kato–Temple theorem. The second concerns small eigenvalues. An absolute bound of size  $\|\mathbf{E}\|_2$  says nothing about an eigenvalue much smaller than  $\|\mathbf{E}\|_2$ , and for perturbations of the form  $\mathbf{A} \mapsto \mathbf{SAS}^*$  we prove a **relative** bound instead, Ostrowski’s theorem, which is Sylvester’s law of inertia made quantitative.

Throughout,  $F = \mathbb{C}$ , matrices whose eigenvalues are ordered are Hermitian, and  $\lambda_1(\mathbf{A}) \geq \dots \geq \lambda_n(\mathbf{A})$  are the eigenvalues of  $\mathbf{A}$  in decreasing order, with multiplicity. Vectors are measured in  $\|\cdot\|_2$ , written  $\|\cdot\|$  when no other norm is in sight, and  $\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{y}^* \mathbf{x}$ .

### 19.6.1 What Weyl’s bound leaves open

Recall Corollary 16.3.6: if  $\mathbf{A}$  and  $\mathbf{E}$  are Hermitian, then  $|\lambda_i(\mathbf{A} + \mathbf{E}) - \lambda_i(\mathbf{A})| \leq \|\mathbf{E}\|_2$  for every  $i$ . By Corollary 16.3.7 the constant 1 cannot be lowered, since  $\mathbf{E} = t\mathbf{I}$  moves every eigenvalue by exactly  $t = \|\mathbf{E}\|_2$ . As a bound on **absolute** error, for an **unknown** Hermitian perturbation of known size, there is nothing more to be said.

The two assumptions in bold are where there is room. First, in practice  $\mathbf{E}$  is not known at all. An iterative method stops with a unit vector  $\mathbf{x}$  and a number  $\mu$  that it believes to be an eigenpair, and what can be measured is the **residual**  $\mathbf{r} = \mathbf{Ax} - \mu\mathbf{x}$ . Second, an absolute error bound is only informative for eigenvalues larger than the error. For the matrix  $\text{diag}(1, 10^{-6})$  and a Hermitian perturbation of size  $10^{-2}$ , Weyl’s bound allows the eigenvalue  $10^{-6}$  to end up anywhere in an interval of radius  $10^{-2}$ , including negative numbers. Whether that is a fair description depends on the kind of perturbation, and for an important kind it is not.

### 19.6.2 Residual bounds

Suppose  $\mathbf{x}$  is a unit vector and  $\mu$  a real number, and  $\mathbf{r} = \mathbf{Ax} - \mu\mathbf{x}$  is small. If  $\mathbf{r} = \mathbf{0}$ , then  $\mu$  is an eigenvalue. The next theorem says that if  $\mathbf{r}$  is merely small,  $\mu$  is **near** an eigenvalue, with the size of  $\mathbf{r}$  as the distance.

#### Theorem 19.6.1 — Residual Bound for Hermitian Matrices

Let  $\mathbf{A} \in M_n(\mathbb{C})$  be Hermitian, let  $\mathbf{x} \in \mathbb{C}^n$  with  $\|\mathbf{x}\| = 1$ , let  $\mu \in \mathbb{R}$ , and put  $\mathbf{r} = \mathbf{Ax} - \mu\mathbf{x}$ . Then some eigenvalue  $\lambda$  of  $\mathbf{A}$  satisfies  $|\lambda - \mu| \leq \|\mathbf{r}\|$ ; that is, the interval  $[\mu - \|\mathbf{r}\|, \mu + \|\mathbf{r}\|]$  contains an eigenvalue of  $\mathbf{A}$ .

This is the normal case of §04’s Theorem 19.4.4, since a Hermitian matrix is normal. We give the direct proof from the spectral theorem instead, because the expansion it uses is the one Kato–Temple needs.

*Proof.* By Corollary 11.4.3 there is an orthonormal basis  $(\mathbf{q}_1, \dots, \mathbf{q}_n)$  of  $\mathbb{C}^n$  with  $\mathbf{A}\mathbf{q}_i = \lambda_i \mathbf{q}_i$ , where  $\lambda_i = \lambda_i(\mathbf{A})$ . Write  $\mathbf{x} = \sum_i c_i \mathbf{q}_i$ , with  $\sum_i |c_i|^2 = \|\mathbf{x}\|^2 = 1$  by Theorem 10.2.1 (c). Then  $\mathbf{r} = \sum_i c_i (\lambda_i - \mu) \mathbf{q}_i$ , and by the same result

$$\|\mathbf{r}\|^2 = \sum_{i=1}^n |c_i|^2 (\lambda_i - \mu)^2 \geq \left( \min_i (\lambda_i - \mu)^2 \right) \sum_{i=1}^n |c_i|^2 = \min_i (\lambda_i - \mu)^2.$$

Taking square roots,  $\min_i |\lambda_i - \mu| \leq \|\mathbf{r}\|$ , as claimed. ■

The bound holds for every real  $\mu$ , and one choice of  $\mu$  makes the residual smallest: the **Rayleigh quotient**  $R_{\mathbf{A}}(\mathbf{x}) = \mathbf{x}^* \mathbf{A} \mathbf{x}$  of Definition 16.1.1, which for a unit vector needs no denominator. With that choice the residual is orthogonal to  $\mathbf{x}$ , since

$$\langle \mathbf{r}, \mathbf{x} \rangle = \mathbf{x}^* \mathbf{A} \mathbf{x} - \mu \mathbf{x}^* \mathbf{x} = \mu - \mu = 0,$$

and Pythagoras (Theorem 10.1.3) applied to  $\mathbf{A}\mathbf{x} = \mu\mathbf{x} + \mathbf{r}$  gives

$$\|\mathbf{r}\|^2 = \|\mathbf{A}\mathbf{x}\|^2 - \mu^2 \quad (\mu = R_{\mathbf{A}}(\mathbf{x}), \|\mathbf{x}\| = 1). \quad (19.6.1)$$

Exercise B3 shows that any other  $\mu$  gives a larger residual. From now on,  $\mu$  is the Rayleigh quotient and  $\varepsilon = \|\mathbf{A}\mathbf{x} - \mu\mathbf{x}\|$  its residual.

*Quick check.* Let  $\mathbf{A} = \text{diag}(1, 3)$  and  $\mathbf{x} = \frac{1}{\sqrt{2}}(1, 1)$ . Compute  $\mu = R_{\mathbf{A}}(\mathbf{x})$ , the residual  $\mathbf{r}$  and  $\varepsilon = \|\mathbf{r}\|$ . Which eigenvalues does the interval  $[\mu - \varepsilon, \mu + \varepsilon]$  contain?

*Solution.*  $\mathbf{A}\mathbf{x} = \frac{1}{\sqrt{2}}(1, 3)$ , so  $\mu = \mathbf{x}^* \mathbf{A} \mathbf{x} = \frac{1}{2}(1 + 3) = 2$  and  $\mathbf{r} = \frac{1}{\sqrt{2}}(1 - 2, 3 - 2) = \frac{1}{\sqrt{2}}(-1, 1)$ , with  $\varepsilon = 1$ . As a check, Eq. 19.6.1 gives  $\|\mathbf{A}\mathbf{x}\|^2 - \mu^2 = 5 - 4 = 1$ . The interval is  $[1, 3]$ , and it contains both eigenvalues, at its two ends. The bound is attained: a vector halfway between two eigenvectors is as bad an approximation as it can be.

### 19.6.3 The Kato–Temple bound

The residual bound is linear in  $\varepsilon$ , and in general no better is possible: in the Quick check,  $|\lambda - \mu| = \varepsilon$  for both eigenvalues. But that example had  $\mathbf{x}$  equally far from both eigenvectors. When  $\mathbf{x}$  is close to one eigenvector, the Rayleigh quotient is a much better estimate than  $\varepsilon$  suggests. Exercise (C1: Why the quotient estimates eigenvalues so well) of Chapter 16 found that an eigenvector error of size  $\delta$  costs only about  $\delta^2$  in the Rayleigh quotient. The trouble is that the eigenvector error cannot be measured, while the residual can. The next theorem restates the  $\delta^2$  phenomenon in terms of the residual, and the price is a hypothesis on how far the other eigenvalues are.

Here is where the gap comes in. If no eigenvalue of  $\mathbf{A}$  lies strictly between two real numbers  $\lambda$  and  $\beta$ , then  $(t - \lambda)(t - \beta) \geq 0$  at every eigenvalue  $t$ , so the matrix  $(\mathbf{A} - \lambda\mathbf{I})(\mathbf{A} - \beta\mathbf{I})$  is positive semidefinite. Evaluating its quadratic form at  $\mathbf{x}$  gives an inequality that involves only  $\mu$ ,  $\varepsilon$ ,  $\lambda$  and  $\beta$ .

**Theorem 19.6.2 — Kato–Temple Theorem**

Let  $\mathbf{A} \in M_n(\mathbb{C})$  be Hermitian, let  $\mathbf{x} \in \mathbb{C}^n$  with  $\|\mathbf{x}\| = 1$ , let  $\mu = \mathbf{x}^* \mathbf{A} \mathbf{x}$ , and let  $\varepsilon = \|\mathbf{A} \mathbf{x} - \mu \mathbf{x}\|$ .

- (a) Let  $\beta > \mu$ , and let  $\lambda < \beta$  be an eigenvalue of  $\mathbf{A}$  such that **no** eigenvalue of  $\mathbf{A}$  lies in the open interval  $(\lambda, \beta)$ . Then

$$\lambda \geq \mu - \frac{\varepsilon^2}{\beta - \mu}.$$

- (b) Let  $\alpha < \mu$ , and let  $\lambda > \alpha$  be an eigenvalue of  $\mathbf{A}$  such that **no** eigenvalue of  $\mathbf{A}$  lies in the open interval  $(\alpha, \lambda)$ . Then

$$\lambda \leq \mu + \frac{\varepsilon^2}{\mu - \alpha}.$$

In particular, if  $\alpha < \mu < \beta$  and  $\lambda$  is the **only** eigenvalue of  $\mathbf{A}$  in  $(\alpha, \beta)$ , then

$$\mu - \frac{\varepsilon^2}{\beta - \mu} \leq \lambda \leq \mu + \frac{\varepsilon^2}{\mu - \alpha}.$$

*Idea.* Compute the number  $\mathbf{x}^*(\mathbf{A} - \lambda \mathbf{I})(\mathbf{A} - \beta \mathbf{I})\mathbf{x}$  two ways. ① In the eigenbasis it is  $\sum_i |c_i|^2 (\lambda_i - \lambda)(\lambda_i - \beta)$ , and the gap hypothesis makes every term non-negative: each eigenvalue lies on or outside the interval between  $\lambda$  and  $\beta$ . ② Expanding the product, it is  $\|\mathbf{A} \mathbf{x}\|^2 - (\lambda + \beta)\mu + \lambda\beta$ , and [Eq. 19.6.1](#) turns this into  $\varepsilon^2 + (\mu - \lambda)(\mu - \beta)$ . So  $(\mu - \lambda)(\beta - \mu) \leq \varepsilon^2$ ; divide by the positive number  $\beta - \mu$ . Part (b) is the same with  $\alpha$  in place of  $\beta$ .

*Proof.*

- (a) As in the proof of [Theorem 19.6.1](#), write  $\mathbf{x} = \sum_i c_i \mathbf{q}_i$  in an orthonormal eigenbasis,  $\mathbf{A} \mathbf{q}_i = \lambda_i \mathbf{q}_i$ ,  $\sum_i |c_i|^2 = 1$ . Then  $(\mathbf{A} - \lambda \mathbf{I})(\mathbf{A} - \beta \mathbf{I})\mathbf{x} = \sum_i c_i (\lambda_i - \lambda)(\lambda_i - \beta) \mathbf{q}_i$ , and by [Theorem 10.2.1 \(b\)](#),

$$\mathbf{x}^*(\mathbf{A} - \lambda \mathbf{I})(\mathbf{A} - \beta \mathbf{I})\mathbf{x} = \sum_{i=1}^n |c_i|^2 (\lambda_i - \lambda)(\lambda_i - \beta).$$

Each eigenvalue  $\lambda_i$  satisfies  $\lambda_i \leq \lambda$  or  $\lambda_i \geq \beta$ , because none lies in  $(\lambda, \beta)$ . In the first case both factors are  $\leq 0$ , in the second both are  $\geq 0$ , since  $\lambda < \beta$ . So every term is  $\geq 0$ , and the left-hand side is  $\geq 0$ .

On the other hand, expanding the product and using  $\mathbf{x}^* \mathbf{x} = 1$ ,  $\mathbf{x}^* \mathbf{A} \mathbf{x} = \mu$  and  $\mathbf{x}^* \mathbf{A}^2 \mathbf{x} = (\mathbf{A} \mathbf{x})^* (\mathbf{A} \mathbf{x}) = \|\mathbf{A} \mathbf{x}\|^2$ , where  $\mathbf{A}^* = \mathbf{A}$ ,

$$\begin{aligned} \mathbf{x}^*(\mathbf{A} - \lambda \mathbf{I})(\mathbf{A} - \beta \mathbf{I})\mathbf{x} &= \|\mathbf{A} \mathbf{x}\|^2 - (\lambda + \beta)\mu + \lambda\beta \\ &= \varepsilon^2 + \mu^2 - (\lambda + \beta)\mu + \lambda\beta \\ &= \varepsilon^2 + (\mu - \lambda)(\mu - \beta), \end{aligned}$$

where the second line is [Eq. 19.6.1](#). Hence  $\varepsilon^2 + (\mu - \lambda)(\mu - \beta) \geq 0$ , that is,  $(\mu - \lambda)(\beta - \mu) \leq \varepsilon^2$ . Since  $\beta - \mu > 0$ , dividing gives  $\mu - \lambda \leq \varepsilon^2/(\beta - \mu)$ , which is (a).

- (b) Apply (a) to  $-\mathbf{A}$ , the unit vector  $\mathbf{x}$ , the eigenvalue  $-\lambda$  and the number  $-\alpha$ . The Rayleigh quotient of  $-\mathbf{A}$  at  $\mathbf{x}$  is  $-\mu$ , its residual is  $-(\mathbf{A}\mathbf{x} - \mu\mathbf{x})$ , of the same norm  $\varepsilon$ , and  $-\alpha > -\mu$ . The eigenvalues of  $-\mathbf{A}$  are the negatives of those of  $\mathbf{A}$ , so  $-\lambda < -\alpha$  and none lies in  $(-\lambda, -\alpha)$ . Part (a) gives  $-\lambda \geq -\mu - \varepsilon^2/(-\alpha + \mu)$ , which is (b).

The final statement follows: if  $\lambda$  is the only eigenvalue in  $(\alpha, \beta)$ , then no eigenvalue lies in  $(\lambda, \beta)$  or in  $(\alpha, \lambda)$ , so (a) and (b) both apply. This proves the theorem. ■

Compare the two bounds on the error  $|\lambda - \mu|$ . The residual bound gives  $\varepsilon$ . Kato–Temple gives  $\varepsilon^2$  divided by the distance from  $\mu$  to the neighboring part of the spectrum. When  $\varepsilon$  is small compared with that gap, the square wins by a wide margin: a residual of  $10^{-4}$  and a gap of 1 give an error of at most  $10^{-8}$ . *The Rayleigh quotient's error is quadratic in the residual, provided there is a gap.* That is the reason the Rayleigh quotient is the eigenvalue estimate of choice, and Chapter 23 builds an iteration on it.

The theorem needs a number  $\alpha$  or  $\beta$  that is known to lie beyond the neighboring eigenvalue, and in practice that information has to come from somewhere. The next example gets it from interlacing.

#### Example. (Bracketing the top eigenvalue)

Let

$$\mathbf{A} = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix}, \quad \mathbf{x} = \frac{1}{\sqrt{3}}(1, 1, 1).$$

Estimate  $\lambda_1(\mathbf{A})$  from  $\mathbf{x}$ : compute  $\mu$  and  $\varepsilon$ , the interval given by [Theorem 19.6.1](#), and the interval given by [Theorem 19.6.2](#), using only information that can be read off  $\mathbf{A}$  without computing its eigenvalues. Compare with the true value  $\lambda_1(\mathbf{A}) = 2 + \sqrt{2}$ .

*Solution. Rayleigh quotient and residual.*  $\mathbf{A}(1, 1, 1) = (3, 4, 3)$ , so  $\mathbf{A}\mathbf{x} = \frac{1}{\sqrt{3}}(3, 4, 3)$  and

$$\mu = \mathbf{x}^* \mathbf{A} \mathbf{x} = \frac{1}{3}(3 + 4 + 3) = \frac{10}{3}, \quad \|\mathbf{A}\mathbf{x}\|^2 = \frac{1}{3}(9 + 16 + 9) = \frac{34}{3}.$$

By [Eq. 19.6.1](#),  $\varepsilon^2 = \frac{34}{3} - \frac{100}{9} = \frac{2}{9}$ , so  $\varepsilon = \frac{\sqrt{2}}{3} \approx 0.471$ . Directly:  $\mathbf{r} = \frac{1}{\sqrt{3}}(3 - \frac{10}{3}, 4 - \frac{10}{3}, 3 - \frac{10}{3}) = \frac{1}{3\sqrt{3}}(-1, 2, -1)$ , of squared norm  $\frac{6}{27} = \frac{2}{9}$ .

*Residual bound.* Some eigenvalue lies in  $[\frac{10}{3} - 0.471, \frac{10}{3} + 0.471] \approx [2.862, 3.805]$ .

*Kato–Temple.* By Proposition 16.1.1,  $\lambda_1(\mathbf{A}) \geq \mu = \frac{10}{3}$ . For an upper bound we need  $\alpha < \mu$  with no eigenvalue in  $(\alpha, \lambda_1)$ , which says  $\lambda_2(\mathbf{A}) \leq \alpha$ . Deleting the second row and column leaves  $\mathbf{B} = \text{diag}(2, 2)$ , and Cauchy's interlacing theorem (Theorem 16.4.2, with  $I = \{1, 3\}$ ) gives  $\lambda_2(\mathbf{A}) \leq \lambda_1(\mathbf{B}) = 2$ . So  $\alpha = 2$  is allowed:  $\alpha < \mu$ , and no eigenvalue lies in  $(2, \lambda_1)$ , because  $\lambda_2 \leq 2$  and  $\lambda_1 \geq \frac{10}{3} > 2$ . Part (b) gives

$$\lambda_1(\mathbf{A}) \leq \frac{10}{3} + \frac{2/9}{10/3 - 2} = \frac{10}{3} + \frac{2}{9} \cdot \frac{3}{4} = \frac{10}{3} + \frac{1}{6} = \frac{7}{2}.$$

So  $\frac{10}{3} \leq \lambda_1(\mathbf{A}) \leq \frac{7}{2}$ , an interval of width  $\frac{1}{6} \approx 0.167$ , against width  $2\varepsilon \approx 0.943$  for the residual bound.

*Check.* The true value is  $2 + \sqrt{2} \approx 3.414$ , which lies in both intervals; the other eigenvalues are 2 and  $2 - \sqrt{2}$ . The actual error of the Rayleigh quotient is  $2 + \sqrt{2} - \frac{10}{3} \approx 0.081$ , about a sixth of  $\varepsilon$ .

Kato–Temple cannot be improved in general: for  $\mathbf{A} = \text{diag}(\lambda_1, \lambda_2)$  with  $\lambda_1 > \lambda_2$ , and  $\alpha = \lambda_2$  the other eigenvalue, the upper bound in (b) is an **equality** at every real unit  $\mathbf{x}$  that is not an eigenvector for  $\lambda_2$  (Exercise C3). At such an eigenvector the hypothesis  $\alpha < \mu$  fails, since there  $\mu = \lambda_2$ .

*Warning.* **The gap hypothesis is not decoration.** Take  $\mathbf{A} = \text{diag}(1, 0)$  and  $\mathbf{x} = \frac{1}{\sqrt{2}}(1, 1)$ , so  $\mu = \frac{1}{2}$  and  $\mathbf{r} = \frac{1}{\sqrt{2}}(\frac{1}{2}, -\frac{1}{2})$ ,  $\varepsilon^2 = \frac{1}{4}$ . With  $\lambda = 1$  and  $\alpha = 0$ , part (b) gives  $1 \leq \frac{1}{2} + \frac{1/4}{1/2} = 1$ , true with equality. With  $\alpha = -\frac{1}{2}$  instead, the eigenvalue 0 lies in  $(\alpha, \lambda)$ , the hypothesis fails, and the formula would claim  $1 \leq \frac{1}{2} + \frac{1/4}{1} = \frac{3}{4}$ , which is false. A number  $\alpha$  chosen too far from  $\mu$ , past an unseen eigenvalue, gives a bound that is simply wrong.

## 19.6.4 Relative perturbations and congruence

Now the second question: small eigenvalues. Weyl’s bound is absolute, and an absolute bound is useless for an eigenvalue smaller than itself. Some perturbations deserve better. In a change of variables  $\mathbf{x} = \mathbf{S}^*\mathbf{y}$  with  $\mathbf{S}$  invertible, the quadratic form  $\mathbf{x}^*\mathbf{A}\mathbf{x}$  becomes  $\mathbf{y}^*(\mathbf{S}\mathbf{A}\mathbf{S}^*)\mathbf{y}$ ; the matrix undergoes a **congruence**, as in Chapter 13 and in Chapter 16 §10. When  $\mathbf{S}$  is close to  $\mathbf{I}$ , for instance when it comes from a slightly inaccurate elimination step or from rounding in a factor,  $\mathbf{S}\mathbf{A}\mathbf{S}^*$  is a perturbation of  $\mathbf{A}$ , but a very structured one. Its effect on the eigenvalues is **multiplicative**.

Two facts are known already. By Theorem 16.10.1 (b), applied with  $\mathbf{P} = \mathbf{S}^*$ , a congruence preserves the numbers of positive, negative and zero eigenvalues; it cannot change a sign. And by Chapter 13 §05’s warning, it can still move eigenvalues a long way: with  $\mathbf{P} = \text{diag}(2, 3)$ ,  $\mathbf{P}^\top \text{diag}(1, -1)\mathbf{P} = \text{diag}(4, -9)$ . Ostrowski’s theorem measures how far.

### Theorem 19.6.3 — Ostrowski’s Theorem

Let  $\mathbf{A} \in M_n(\mathbb{C})$  be Hermitian and let  $\mathbf{S} \in M_n(\mathbb{C})$  be invertible. Then  $\mathbf{S}\mathbf{S}^*$  is positive definite, and for each  $k = 1, \dots, n$  there is a real number  $\theta_k$  with

$$\lambda_k(\mathbf{S}\mathbf{A}\mathbf{S}^*) = \theta_k \lambda_k(\mathbf{A}), \quad \lambda_n(\mathbf{S}\mathbf{S}^*) \leq \theta_k \leq \lambda_1(\mathbf{S}\mathbf{S}^*).$$

In particular  $\theta_k > 0$ , and

$$|\lambda_k(\mathbf{S}\mathbf{A}\mathbf{S}^*) - \lambda_k(\mathbf{A})| \leq \|\mathbf{S}\mathbf{S}^* - \mathbf{I}\|_2 |\lambda_k(\mathbf{A})|.$$

*Idea. Step roadmap.* Fix  $k$  and put  $\mu = \lambda_k(\mathbf{A})$ .

① **Shift, so the  $k$ -th eigenvalue is 0.**  $\mathbf{A} - \mu\mathbf{I}$  has at most  $k - 1$  positive eigenvalues and at least  $k$  non-negative ones. A congruence preserves both counts (Sylvester’s law), so  $\mathbf{S}(\mathbf{A} - \mu\mathbf{I})\mathbf{S}^*$  has the same property, and that forces its  $k$ -th eigenvalue to be exactly 0.

② **Split.**  $\mathbf{S}\mathbf{A}\mathbf{S}^* = \mathbf{S}(\mathbf{A} - \mu\mathbf{I})\mathbf{S}^* + \mu\mathbf{S}\mathbf{S}^*$ .

③ **Weyl.** Adding  $\mu\mathbf{S}\mathbf{S}^*$  moves the  $k$ -th eigenvalue from 0 to somewhere between the smallest and the largest eigenvalue of  $\mu\mathbf{S}\mathbf{S}^*$ , which are  $\mu\lambda_n(\mathbf{S}\mathbf{S}^*)$  and  $\mu\lambda_1(\mathbf{S}\mathbf{S}^*)$  in some order, depending on the sign of  $\mu$ .

*Proof.* For  $\mathbf{x} \neq \mathbf{0}$ ,  $\mathbf{x}^*\mathbf{S}\mathbf{S}^*\mathbf{x} = \|\mathbf{S}^*\mathbf{x}\|^2 > 0$ , because  $\mathbf{S}^*$  is invertible. So  $\mathbf{S}\mathbf{S}^*$ , which is Hermitian, is positive definite, and its eigenvalues are positive by Theorem 12.1.5; write  $m = \lambda_n(\mathbf{S}\mathbf{S}^*) > 0$  and  $M = \lambda_1(\mathbf{S}\mathbf{S}^*)$ .

Fix  $k$  and put  $\mu = \lambda_k(\mathbf{A})$ . The eigenvalues of the Hermitian matrix  $\mathbf{A} - \mu\mathbf{I}$  are  $\lambda_i(\mathbf{A}) - \mu$ , in the same decreasing order. Those with  $i \geq k$  are  $\leq 0$ , so at most  $k - 1$  of them are positive; those with  $i \leq k$  are  $\geq 0$ , so at least  $k$  are non-negative. In the notation of Definition 16.10.1,  $n_+(\mathbf{A} - \mu\mathbf{I}) \leq k - 1$  and  $n_+(\mathbf{A} - \mu\mathbf{I}) + n_0(\mathbf{A} - \mu\mathbf{I}) \geq k$ . Put  $\mathbf{Y} = \mathbf{S}(\mathbf{A} - \mu\mathbf{I})\mathbf{S}^*$ . This is  $\mathbf{P}^*(\mathbf{A} - \mu\mathbf{I})\mathbf{P}$  with  $\mathbf{P} = \mathbf{S}^*$  invertible, so by Theorem 16.10.1 (b),  $\mathbf{Y}$  is Hermitian with the same inertia triple. Hence  $\mathbf{Y}$  has at most  $k - 1$  positive eigenvalues, so  $\lambda_k(\mathbf{Y}) \leq 0$ ; and at least  $k$  non-negative ones, so  $\lambda_k(\mathbf{Y}) \geq 0$ . Thus  $\lambda_k(\mathbf{Y}) = 0$ .

Now  $\mathbf{S}\mathbf{A}\mathbf{S}^* = \mathbf{Y} + \mu\mathbf{S}\mathbf{S}^*$ , a sum of two Hermitian matrices. By Corollary 16.3.4,

$$\lambda_n(\mu\mathbf{S}\mathbf{S}^*) \leq \lambda_k(\mathbf{S}\mathbf{A}\mathbf{S}^*) - \lambda_k(\mathbf{Y}) \leq \lambda_1(\mu\mathbf{S}\mathbf{S}^*),$$

and  $\lambda_k(\mathbf{Y}) = 0$ . If  $\mu \geq 0$ , multiplying by  $\mu$  keeps the order of the eigenvalues, so  $\lambda_n(\mu\mathbf{S}\mathbf{S}^*) = \mu m$  and  $\lambda_1(\mu\mathbf{S}\mathbf{S}^*) = \mu M$ . If  $\mu < 0$ , then  $\mu\mathbf{S}\mathbf{S}^* = -(|\mu|\mathbf{S}\mathbf{S}^*)$ , and by Lemma 16.3.2 its extreme eigenvalues are  $\lambda_n = -|\mu|M = \mu M$  and  $\lambda_1 = -|\mu|m = \mu m$ . In both cases  $\lambda_k(\mathbf{S}\mathbf{A}\mathbf{S}^*)$  lies between  $\mu m$  and  $\mu M$ . If  $\mu \neq 0$ , put  $\theta_k = \lambda_k(\mathbf{S}\mathbf{A}\mathbf{S}^*)/\mu$ ; dividing by  $\mu$  (and reversing the inequalities when  $\mu < 0$ ) gives  $m \leq \theta_k \leq M$ . If  $\mu = 0$ , both ends are 0, so  $\lambda_k(\mathbf{S}\mathbf{A}\mathbf{S}^*) = 0$ , and  $\theta_k = m$  will do. In either case  $\theta_k \geq m > 0$ .

For the last inequality,  $|\lambda_k(\mathbf{S}\mathbf{A}\mathbf{S}^*) - \lambda_k(\mathbf{A})| = |\theta_k - 1||\mu|$ , and  $\theta_k - 1$  lies in  $[m - 1, M - 1]$ . The numbers  $m - 1$  and  $M - 1$  are the smallest and largest eigenvalues of the Hermitian matrix  $\mathbf{S}\mathbf{S}^* - \mathbf{I}$ , so by Lemma 16.3.5 both have absolute value at most  $\|\mathbf{S}\mathbf{S}^* - \mathbf{I}\|_2$ , and so does everything between them. This proves the theorem. ■

This is a quantitative Sylvester's law of inertia. Sylvester's law says that a congruence cannot change the sign of an eigenvalue; Ostrowski's theorem says that it multiplies the  $k$ -th eigenvalue by a factor  $\theta_k$  that is always positive and always between the extreme eigenvalues of  $\mathbf{S}\mathbf{S}^*$ . These are  $\lambda_1(\mathbf{S}\mathbf{S}^*) = \max_{\|\mathbf{y}\|=1} \|\mathbf{S}^*\mathbf{y}\|^2$  and  $\lambda_n(\mathbf{S}\mathbf{S}^*) = \min_{\|\mathbf{y}\|=1} \|\mathbf{S}^*\mathbf{y}\|^2$ , by Proposition 16.1.1, since  $\mathbf{y}^*\mathbf{S}\mathbf{S}^*\mathbf{y} = \|\mathbf{S}^*\mathbf{y}\|^2$ : the most and the least that the change of variables stretches a vector, squared. When  $\mathbf{S}$  is close to a unitary matrix, both are close to 1, and **every** eigenvalue, however small, keeps most of its significant digits.

*Quick check.* Chapter 13 §05's warning used  $\mathbf{A} = \text{diag}(1, -1)$  and  $\mathbf{P} = \text{diag}(2, 3)$ , with  $\mathbf{P}^\top \mathbf{A} \mathbf{P} = \text{diag}(4, -9)$ . Take  $\mathbf{S} = \mathbf{P}^\top = \mathbf{P}$ . Find  $\theta_1, \theta_2$  and the interval  $[\lambda_n(\mathbf{S}\mathbf{S}^*), \lambda_1(\mathbf{S}\mathbf{S}^*)]$ . Can the interval in Theorem 19.6.3 be shortened?

*Solution.*  $\mathbf{S}\mathbf{S}^* = \text{diag}(4, 9)$ , so the interval is  $[4, 9]$ . The eigenvalues of  $\mathbf{S}\mathbf{A}\mathbf{S}^* = \text{diag}(4, -9)$  in decreasing order are 4 and  $-9$ , and those of  $\mathbf{A}$  are 1 and  $-1$ . So  $\theta_1 = 4/1 = 4$  and  $\theta_2 = -9/(-1) = 9$ . They sit at the two ends of the interval, so neither end can be moved inwards: the interval is sharp.

The next example shows the difference between the absolute and the relative view on a matrix whose eigenvalues differ greatly in size.

**Example. (A tiny eigenvalue under a change of variables)**

Let  $\mathbf{A} = \text{diag}(1, 10^{-6})$  and  $\mathbf{S} = \begin{pmatrix} 1 & 0 \\ t & 1 \end{pmatrix}$  with  $t = 10^{-2}$ , the change of variables that adds  $t$  times the first coordinate to the second. Compute  $\mathbf{SAS}^*$  and  $\mathbf{E} = \mathbf{SAS}^* - \mathbf{A}$ . What do Corollary 16.3.4 and Theorem 19.6.3 say about  $\lambda_2(\mathbf{SAS}^*)$ , and what is its true value?

*Solution.* Write  $\eta = 10^{-6}$ . Multiplying out,

$$\mathbf{SAS}^* = \begin{pmatrix} 1 & t \\ t & t^2 + \eta \end{pmatrix}, \quad \mathbf{E} = \begin{pmatrix} 0 & t \\ t & t^2 \end{pmatrix},$$

with  $t^2 + \eta = 1.01 \times 10^{-4}$ .

*Weyl.*  $\mathbf{E}$  has trace  $t^2$  and determinant  $-t^2$ , so its eigenvalues are  $\frac{1}{2}(t^2 \pm t\sqrt{t^2 + 4})$ , approximately 0.01005 and  $-0.00995$ . By Corollary 16.3.4,

$$10^{-6} - 0.00995 \leq \lambda_2(\mathbf{SAS}^*) \leq 10^{-6} + 0.01005.$$

This interval is about  $10^4$  times wider than the eigenvalue it is meant to locate, and it contains negative numbers.

*Ostrowski.*  $\mathbf{SS}^* = \begin{pmatrix} 1 & t \\ t & 1+t^2 \end{pmatrix}$  has trace  $2 + t^2$  and determinant 1, so its eigenvalues are  $\frac{1}{2}(2 + t^2 \pm t\sqrt{4 + t^2})$ , approximately 1.01005 and 0.99005. By Theorem 19.6.3,

$$0.99005 \times 10^{-6} \leq \lambda_2(\mathbf{SAS}^*) \leq 1.01005 \times 10^{-6} :$$

the eigenvalue is positive and known to within about 1%.

*The truth.*  $\det(\mathbf{SAS}^*) = \det(\mathbf{S})^2 \det \mathbf{A} = \eta$ , since  $\det \mathbf{S} = 1$ , and the trace is  $1 + t^2 + \eta$ . So the two eigenvalues multiply to  $\eta$ , and the larger is  $1.0001\dots$ ; hence  $\lambda_2(\mathbf{SAS}^*) = \eta/\lambda_1(\mathbf{SAS}^*) \approx 0.99990 \times 10^{-6}$ . The relative change is about  $10^{-4}$ , well inside Ostrowski's 1%; Weyl's bound, by contrast, cannot even decide the sign.

*Warning.* **Weyl's bound is sharp and still useless for small eigenvalues.** It is the best possible bound on **absolute** error for **arbitrary** Hermitian  $\mathbf{E}$  of a given norm, and for exactly that reason it cannot distinguish an eigenvalue of size  $10^{-6}$  from 0 once  $\|\mathbf{E}\|_2 \geq 10^{-6}$ . A general Hermitian  $\mathbf{E}$  really can destroy a small eigenvalue:  $\mathbf{E} = -10^{-6}\mathbf{e}_2\mathbf{e}_2^\top$  sends the eigenvalue  $10^{-6}$  of  $\text{diag}(1, 10^{-6})$  to 0. Relative accuracy needs a structured perturbation, such as a congruence  $\mathbf{SAS}^*$  with  $\mathbf{S}$  near  $\mathbf{I}$ .

## 19.6.5 Exercises

### 19.6.5.1 A. Check your understanding

**Exercise A1.**

- (a) State the Kato–Temple theorem, with its hypotheses.
- (b) True or false: for Hermitian  $\mathbf{A}$ , a unit vector  $\mathbf{x}$  and real  $\mu$ , the eigenvalue of  $\mathbf{A}$

**closest** to  $\mu$  is within  $\|\mathbf{Ax} - \mu\mathbf{x}\|$  of  $\mu$ . Justify your answer.

- (c) True or false: for Hermitian  $\mathbf{A}$  and invertible  $\mathbf{S}$ ,  $\lambda_k(\mathbf{SAS}^*)$  and  $\lambda_k(\mathbf{A})$  have the same sign for every  $k$ . Justify your answer.
- (d) Which of the three bounds of this section need information that is not contained in  $\mathbf{A}$ ,  $\mathbf{x}$  and  $\mu$ ? What information?
- (e) Why can the constant 1 in Corollary 16.3.6 not be lowered, and why does that not contradict Ostrowski's theorem?

*Solution.*

- (a) Let  $\mathbf{A}$  be Hermitian,  $\mathbf{x}$  a unit vector,  $\mu = \mathbf{x}^*\mathbf{Ax}$ ,  $\varepsilon = \|\mathbf{Ax} - \mu\mathbf{x}\|$ . If  $\beta > \mu$  and  $\lambda < \beta$  is an eigenvalue with no eigenvalue in  $(\lambda, \beta)$ , then  $\lambda \geq \mu - \varepsilon^2/(\beta - \mu)$ . If  $\alpha < \mu$  and  $\lambda > \alpha$  is an eigenvalue with no eigenvalue in  $(\alpha, \lambda)$ , then  $\lambda \leq \mu + \varepsilon^2/(\mu - \alpha)$ .
- (b) True. [Theorem 19.6.1](#) gives some eigenvalue within  $\|\mathbf{Ax} - \mu\mathbf{x}\|$  of  $\mu$ , and the closest one is at least as close.
- (c) True. By [Theorem 19.6.3](#),  $\lambda_k(\mathbf{SAS}^*) = \theta_k \lambda_k(\mathbf{A})$  with  $\theta_k > 0$ ; a positive multiple has the same sign, and 0 goes to 0.
- (d) The residual bound needs nothing more. Kato–Temple needs a number  $\alpha$  or  $\beta$  beyond the neighboring eigenvalue, that is, a lower bound on the gap; in [Example \(Bracketing the top eigenvalue\)](#) it came from interlacing. Ostrowski's theorem is not about an approximate eigenvector, but it needs  $\mathbf{S}$ , or at least the extreme eigenvalues of  $\mathbf{SS}^*$ .
- (e)  $\mathbf{E} = t\mathbf{I}$  moves every eigenvalue by exactly  $t = \|\mathbf{E}\|_2$ . Ostrowski's theorem is about the perturbations  $\mathbf{SAS}^* - \mathbf{A}$ , which form a special class;  $\mathbf{A} + t\mathbf{I}$  is not of the form  $\mathbf{SAS}^*$  in general (for  $\mathbf{A} = \mathbf{0}$  it never is, when  $t \neq 0$ ). A better bound for a smaller class contradicts nothing.

### 19.6.5.2 B. Practice

#### Exercise B1: Three bounds for one estimate.

Let  $\mathbf{A} = \begin{pmatrix} 5 & 2 \\ 2 & 2 \end{pmatrix}$  and  $\mathbf{x} = \frac{1}{\sqrt{10}}(3, 1)$ .

- (a) Compute  $\mu = \mathbf{x}^*\mathbf{Ax}$  and  $\varepsilon = \|\mathbf{Ax} - \mu\mathbf{x}\|$ , and the interval given by [Theorem 19.6.1](#).
- (b) Using Theorem 16.4.2 to bound  $\lambda_2(\mathbf{A})$ , apply [Theorem 19.6.2](#) to bracket  $\lambda_1(\mathbf{A})$ .
- (c) Find the eigenvalues of  $\mathbf{A}$  exactly. Hence compare the actual error of  $\mu$  with  $\varepsilon$  and with the Kato–Temple bound.

*Solution.*

- (a)  $\mathbf{A}(3, 1) = (17, 8)$ , so  $\mu = \frac{1}{10}(3 \cdot 17 + 1 \cdot 8) = \frac{59}{10} = 5.9$ . The residual is  $\frac{1}{\sqrt{10}}((17, 8) - 5.9(3, 1)) = \frac{1}{\sqrt{10}}(-0.7, 2.1)$ , so  $\varepsilon^2 = \frac{1}{10}(0.49 + 4.41) = 0.49$  and  $\varepsilon = 0.7$ . Some eigenvalue lies in  $[5.2, 6.6]$ .
- (b) Deleting the first row and column leaves  $(2)$ , so Theorem 16.4.2 gives  $\lambda_2(\mathbf{A}) \leq 2$ . Also  $\lambda_1(\mathbf{A}) \geq \mu = 5.9$  by Proposition 16.1.1. So with  $\alpha = 2 < \mu$ , no eigenvalue

lies in  $(2, \lambda_1)$ , and part (b) of [Theorem 19.6.2](#) gives

$$\lambda_1(\mathbf{A}) \leq 5.9 + \frac{0.49}{5.9 - 2} = 5.9 + \frac{0.49}{3.9} \approx 6.026.$$

Hence  $5.9 \leq \lambda_1(\mathbf{A}) \leq 6.026$ .

- (c)  $\mathbf{A}$  has trace 7 and determinant 6, so its eigenvalues are 6 and 1; indeed  $\mathbf{A}(2, 1) = (12, 6)$ . The actual error of  $\mu$  is  $6 - 5.9 = 0.1$ . The residual bound allows 0.7, seven times too much; Kato–Temple allows 0.126, within about 26% of the truth.

### Exercise B2: Ostrowski's factors.

Let  $\mathbf{A} = \text{diag}(2, -1)$  and  $\mathbf{S} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ . Compute  $\mathbf{SAS}^*$  and its eigenvalues, the eigenvalues of  $\mathbf{SS}^*$ , and the factors  $\theta_1, \theta_2$  of [Theorem 19.6.3](#). Hence verify the theorem for this pair.

*Solution.*  $\mathbf{AS}^* = \text{diag}(2, -1) \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ -1 & -1 \end{pmatrix}$ , and

$$\mathbf{SAS}^* = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ -1 & -1 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ -1 & -1 \end{pmatrix},$$

with trace 0 and determinant  $-2$ , so eigenvalues  $\pm\sqrt{2}$ . Next  $\mathbf{SS}^* = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$ , with trace 3 and determinant 1, so eigenvalues  $\frac{3 \pm \sqrt{5}}{2} \approx 2.618$  and  $0.382$ .

Then  $\theta_1 = \lambda_1(\mathbf{SAS}^*)/\lambda_1(\mathbf{A}) = \sqrt{2}/2 \approx 0.707$  and  $\theta_2 = (-\sqrt{2})/(-1) = \sqrt{2} \approx 1.414$ . Both lie in  $[0.382, 2.618]$ , as the theorem says, and both are positive, so the signs  $+, -$  are preserved.

### Exercise B3: The Rayleigh quotient minimizes the residual.

Let  $\mathbf{A}$  be Hermitian and  $\mathbf{x}$  a unit vector, with  $\rho = \mathbf{x}^* \mathbf{A} \mathbf{x}$ . Prove that for every real  $\nu$ ,

$$\|\mathbf{Ax} - \nu \mathbf{x}\|^2 = \|\mathbf{Ax} - \rho \mathbf{x}\|^2 + (\nu - \rho)^2.$$

Hence deduce that among all real  $\nu$ , the residual bound of [Theorem 19.6.1](#) is sharpest at  $\nu = \rho$ .

*Solution.* Write  $\mathbf{Ax} - \nu \mathbf{x} = (\mathbf{Ax} - \rho \mathbf{x}) + (\rho - \nu)\mathbf{x}$ . The first vector is orthogonal to  $\mathbf{x}$ , since  $\langle \mathbf{Ax} - \rho \mathbf{x}, \mathbf{x} \rangle = \mathbf{x}^* \mathbf{Ax} - \rho = 0$ . By [Theorem 10.1.3](#),

$$\|\mathbf{Ax} - \nu \mathbf{x}\|^2 = \|\mathbf{Ax} - \rho \mathbf{x}\|^2 + (\rho - \nu)^2 \|\mathbf{x}\|^2 = \|\mathbf{Ax} - \rho \mathbf{x}\|^2 + (\nu - \rho)^2.$$

Hence  $\|\mathbf{Ax} - \nu \mathbf{x}\| \geq \|\mathbf{Ax} - \rho \mathbf{x}\|$ , with equality only for  $\nu = \rho$ . The interval  $[\nu - \|\mathbf{Ax} - \nu \mathbf{x}\|, \nu + \|\mathbf{Ax} - \nu \mathbf{x}\|]$  of [Theorem 19.6.1](#) therefore has its smallest radius at  $\nu = \rho$ .

### 19.6.5.3 C. Going deeper

#### Exercise C1: A residual that guarantees an eigenvalue.

Let  $\mathbf{A}$  be Hermitian,  $\mathbf{x}$  a unit vector,  $\mu = \mathbf{x}^* \mathbf{A} \mathbf{x}$  and  $\varepsilon = \|\mathbf{Ax} - \mu \mathbf{x}\|$ .

- (a) Let  $\alpha < \mu < \beta$ . Prove that if  $\varepsilon^2 < (\mu - \alpha)(\beta - \mu)$ , then  $\mathbf{A}$  has an eigenvalue in  $(\alpha, \beta)$ .
- (b) Deduce that  $\mathbf{A}$  has an eigenvalue in  $[\mu - \varepsilon, \mu + \varepsilon]$ , which is [Theorem 19.6.1](#) again, for this choice of  $\mu$ .

*Hint for (a): if there is none, consider  $\mathbf{x}^*(\mathbf{A} - \alpha\mathbf{I})(\mathbf{A} - \beta\mathbf{I})\mathbf{x}$ .*

*Solution.*

- (a) Suppose no eigenvalue lies in  $(\alpha, \beta)$ . Then every eigenvalue  $t$  satisfies  $t \leq \alpha$  or  $t \geq \beta$ , so  $(t - \alpha)(t - \beta) \geq 0$ . As in the proof of [Theorem 19.6.2](#), with  $\alpha$  in place of  $\lambda$ ,

$$0 \leq \mathbf{x}^*(\mathbf{A} - \alpha\mathbf{I})(\mathbf{A} - \beta\mathbf{I})\mathbf{x} = \varepsilon^2 + (\mu - \alpha)(\mu - \beta),$$

so  $\varepsilon^2 \geq (\mu - \alpha)(\beta - \mu)$ , contradicting the hypothesis. Hence some eigenvalue lies in  $(\alpha, \beta)$ .

- (b) Let  $d$  be the smallest distance from  $\mu$  to an eigenvalue of  $\mathbf{A}$ , and suppose  $d > \varepsilon$ . Choose  $s$  with  $\varepsilon < s < d$ , and apply (a) with  $\alpha = \mu - s$  and  $\beta = \mu + s$ : the hypothesis holds, since  $(\mu - \alpha)(\beta - \mu) = s^2 > \varepsilon^2$ . So some eigenvalue lies in  $(\mu - s, \mu + s)$ , at distance less than  $s < d$  from  $\mu$ , contradicting the choice of  $d$ . Hence  $d \leq \varepsilon$ , which is the claim.

### Exercise C2: Relative bounds need structured perturbations.

- (a) Let  $\mathbf{A}$  be Hermitian and  $\mathbf{S} = \mathbf{I} + \mathbf{F}$  with  $\mathbf{F} \in M_n(\mathbb{C})$ . Prove that

$$|\lambda_k(\mathbf{SAS}^*) - \lambda_k(\mathbf{A})| \leq (2\|\mathbf{F}\|_2 + \|\mathbf{F}\|_2^2) |\lambda_k(\mathbf{A})|$$

for every  $k$ , whenever  $\mathbf{S}$  is invertible.

- (b) Show that there is **no** constant  $c$  such that  $|\lambda_k(\mathbf{A} + \mathbf{E}) - \lambda_k(\mathbf{A})| \leq c\|\mathbf{E}\|_2|\lambda_k(\mathbf{A})|$  for all Hermitian  $\mathbf{A}$ ,  $\mathbf{E}$  and all  $k$ .

*Hint for (a): expand  $\mathbf{SS}^* - \mathbf{I}$ , and show first that  $\|\mathbf{F}^*\|_2 = \|\mathbf{F}\|_2$ .*

*Solution.*

- (a) First,  $\|\mathbf{F}^*\|_2 \leq \|\mathbf{F}\|_2$ : for every  $\mathbf{x}$ , by [Theorem 10.1.5](#) and [Theorem 15.3.2 \(a\)](#),

$$\|\mathbf{F}^*\mathbf{x}\|^2 = \langle \mathbf{FF}^*\mathbf{x}, \mathbf{x} \rangle \leq \|\mathbf{FF}^*\mathbf{x}\| \|\mathbf{x}\| \leq \|\mathbf{F}\|_2 \|\mathbf{F}^*\mathbf{x}\| \|\mathbf{x}\|,$$

so  $\|\mathbf{F}^*\mathbf{x}\| \leq \|\mathbf{F}\|_2 \|\mathbf{x}\|$  (trivially if  $\mathbf{F}^*\mathbf{x} = \mathbf{0}$ ), and the reverse inequality follows by applying this to  $\mathbf{F}^*$ , since  $(\mathbf{F}^*)^* = \mathbf{F}$ . Now  $\mathbf{SS}^* - \mathbf{I} = \mathbf{F} + \mathbf{F}^* + \mathbf{FF}^*$ , so by the triangle inequality for  $\|\cdot\|_2$ , which is a norm by [Theorem 15.3.2 \(b\)](#), and by part (d) of the same theorem,

$$\|\mathbf{SS}^* - \mathbf{I}\|_2 \leq \|\mathbf{F}\|_2 + \|\mathbf{F}^*\|_2 + \|\mathbf{FF}^*\|_2 = 2\|\mathbf{F}\|_2 + \|\mathbf{F}\|_2^2.$$

The last inequality of [Theorem 19.6.3](#) then gives the claim.

- (b) Take  $\mathbf{A} = \text{diag}(1, 0)$  and  $\mathbf{E} = \text{diag}(0, t)$  with  $t > 0$ . Then  $\lambda_2(\mathbf{A}) = 0$  and  $\lambda_2(\mathbf{A} + \mathbf{E}) = \min(1, t) > 0$ , so the left side is positive while the right side is  $c \cdot t \cdot 0 = 0$ . No

constant  $c$  works. By (a), such an  $\mathbf{E}$  is not of the form  $\mathbf{SAS}^* - \mathbf{A}$ : congruence can never move the eigenvalue 0.

**Exercise C3: Kato–Temple is sharp.**

Let  $\lambda_1 > \lambda_2$  be real,  $\mathbf{A} = \text{diag}(\lambda_1, \lambda_2)$ , and  $\mathbf{x} = (\cos \phi, \sin \phi)$  with  $0 < \phi < \pi/2$ .

- (a) Compute  $\mu = \mathbf{x}^* \mathbf{A} \mathbf{x}$  and show that  $\varepsilon^2 = (\lambda_1 - \lambda_2)^2 \cos^2 \phi \sin^2 \phi$ .
- (b) Show that [Theorem 19.6.2](#) (b), with  $\lambda = \lambda_1$  and  $\alpha = \lambda_2$ , holds with **equality**. Hence the constant in the Kato–Temple bound cannot be improved.

*Solution.* Write  $c = \cos \phi$ ,  $s = \sin \phi$  and  $d = \lambda_1 - \lambda_2 > 0$ .

- (a)  $\mathbf{A} \mathbf{x} = (\lambda_1 c, \lambda_2 s)$ , so  $\mu = \lambda_1 c^2 + \lambda_2 s^2$ . Using  $c^2 + s^2 = 1$ ,  $\lambda_1 - \mu = ds^2$  and  $\lambda_2 - \mu = -dc^2$ . The residual is  $\mathbf{A} \mathbf{x} - \mu \mathbf{x} = ((\lambda_1 - \mu)c, (\lambda_2 - \mu)s) = (ds^2 c, -dc^2 s)$ , so

$$\varepsilon^2 = d^2 s^4 c^2 + d^2 c^4 s^2 = d^2 c^2 s^2 (s^2 + c^2) = d^2 c^2 s^2.$$

- (b) Here  $\alpha = \lambda_2 < \mu$ , since  $\mu - \lambda_2 = dc^2 > 0$ , and no eigenvalue lies in  $(\lambda_2, \lambda_1)$ . The bound reads

$$\mu + \frac{\varepsilon^2}{\mu - \lambda_2} = \mu + \frac{d^2 c^2 s^2}{dc^2} = \mu + ds^2 = \lambda_1.$$

So equality holds for every  $\phi \in (0, \pi/2)$ , and no smaller multiple of  $\varepsilon^2/(\mu - \alpha)$  would give a valid bound.